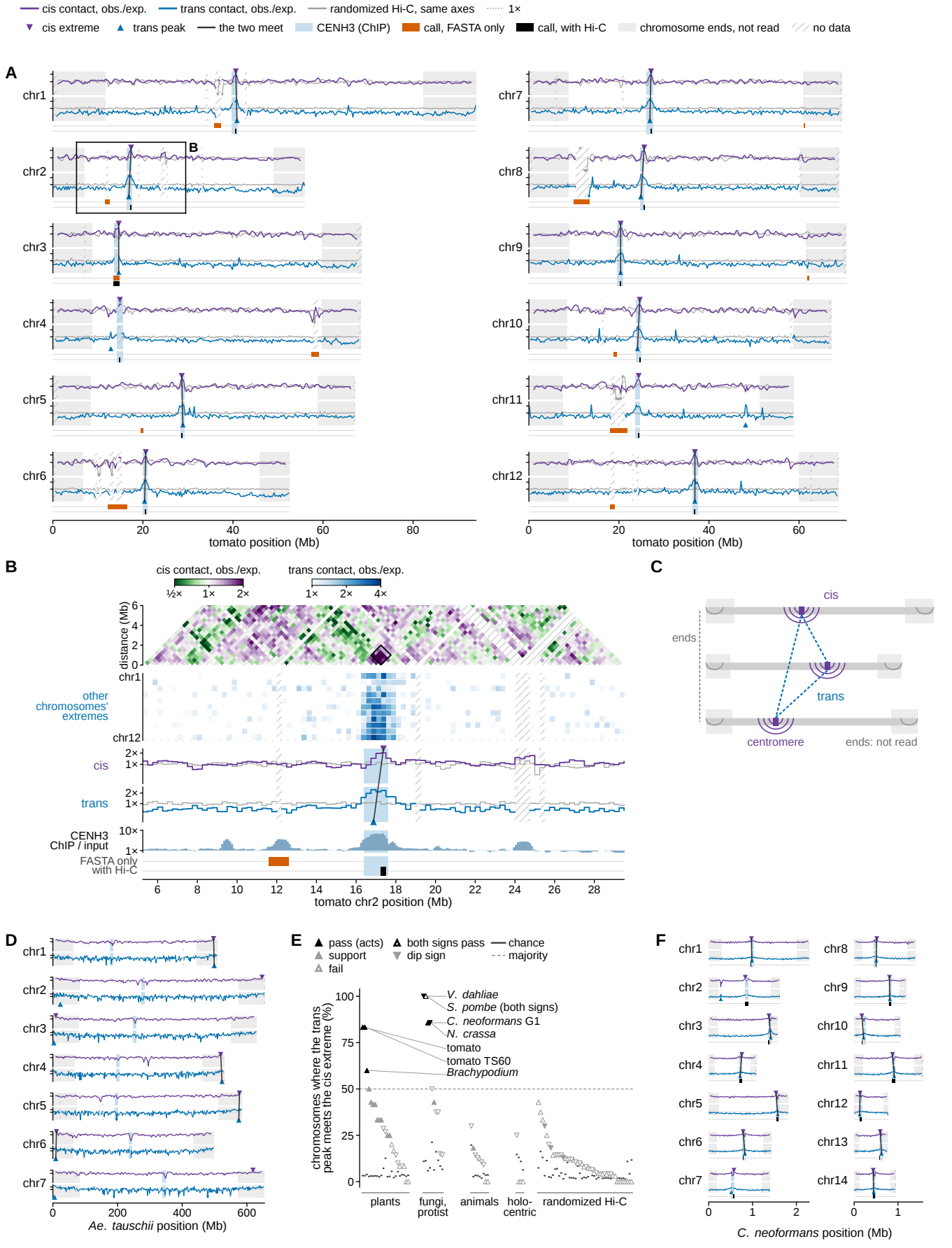

cis contact, obs./exp.
trans contact, obs./exp.
randomized Hi-C, same axes
1×
cis extreme
trans peak
the two meet
CENH3 (ChIP)
call, FASTA only
call, with Hi-C
chromosome ends, not read
no data
A
chr1
chr2
chr3
chr4
chr5
chr6
chr7
chr8
chr9
chr10
chr11
chr12
B
tomato position (Mb)
B
cis contact, obs./exp.
½× 1× 2×
trans contact, obs./exp.
1× 2× 4×
distance (Mb)
other chromosomes' extremes
cis
trans
CENH3 ChIP / input
FASTA only with Hi-C
tomato chr2 position (Mb)
C
cis
ends
trans
centromere
ends: not read
D
Ae. tauschii position (Mb)
E
pass (acts)
support
fail
both signs pass
dip sign
chance
majority
chromosomes where the trans peak meets the cis extreme (%)
V. dahliae
S. pombe (both signs)
C. neoformans G1
N. crassa
tomato
tomato TS60
Brachypodium
plants
fungi, protist
animals
holo-centric
randomized Hi-C
F
C. neoformans position (Mb)

Two readouts of one Hi-C library find centromeres that no repeat marks: where the contact across a stretch of chromosome peaks and the contact with the other chromosomes' peaks rises at the same place, that place is the centromere. No annotation is needed.

A In tomato, the contact across each chromosome's interior peaks once (purple), and the contact with the other chromosomes' peaks (blue) rises at the same place, on CENH3, on 10 of 12 chromosomes (0.39 expected by chance); on chr4 and chr11 the blue maximum sits on a contact hub. A randomized library (gray) shows neither. With Hi-C, all 12 calls land on CENH3 (11 at low confidence), against 1 from the FASTA alone; on chr7 and chr9, whose FASTA calls lay at a chromosome end, the call moves to where that chromosome's own readouts meet (a rule fitted on this genome, so here it is not an independent test). Carets, values beyond the axis.

B On tomato chr2 (boxed in A), a typical chromosome, the two flanks of the centromere contact each other more than expected (diamond), and the same bins contact the centromeric peaks of all 11 other chromosomes (blue rows): both readouts mark one CENH3 domain. The FASTA-only call sits on a satellite array 3.8 Mb away, whose CENH3 signal is 53% of the peak's.

C Schematic: centromeres fold locally (cis) and cluster in the nucleus (trans). Chromosome ends do both too, so only the interior of each chromosome is read.

D In *Ae. tauschii*, read with the ends kept, both readouts peak at chromosome ends and agree there on 4 of 7 chromosomes, away from CENH3; with the ends excluded, none agree.

E Each of 92 Hi-C libraries is one point (bar: the share expected by chance): black where the two readouts meet on more than half of its chromosomes, beyond chance, so the line acts (pass); gray where they meet beyond chance on half or fewer (support); open where they do not beat chance (fail). They agree on most chromosomes, far beyond chance, in tomato (Heinz 1706 and TS60), *Brachypodium*, *C. neoformans*, *N. crassa* and *V. dahliae*; *S. pombe* agrees at both extremes, so no call is made. They agree beyond chance on a minority in 8 of 13 further tomato accessions, barley, *C. neoformans* in anaphase and rhesus. Randomized libraries never agree on a majority (on a minority beyond chance in 3 of 46), and holocentric genomes fail. In potato the dips sit on CENH3 on 9 of 11 chromosomes, but trans contacts do not confirm them.

F In *C. neoformans* (G1 Hi-C, 10-kb bins), the readouts meet on 12 of 14 chromosomes (0.83 expected by chance). With Hi-C, 13 of 14 calls lie on CENP-A at medium confidence and one (chr10) lies 19 kb beyond it; from the FASTA alone all 14 lie on it, at low confidence.

Script: docs/img/src/benchmark_cis_trans.py ·
data: the cenfuse 0.16.0 benchmark runs (not in the repository) · 2026-11-14